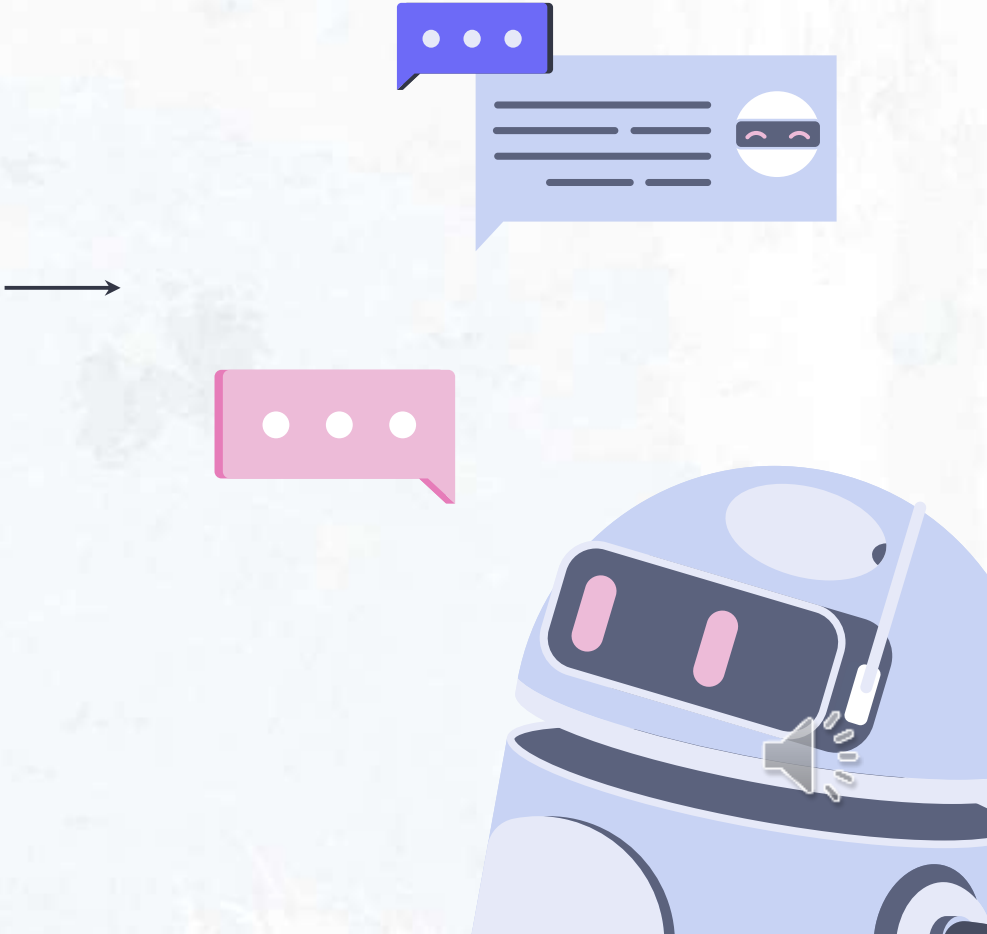
Natural Language Processing



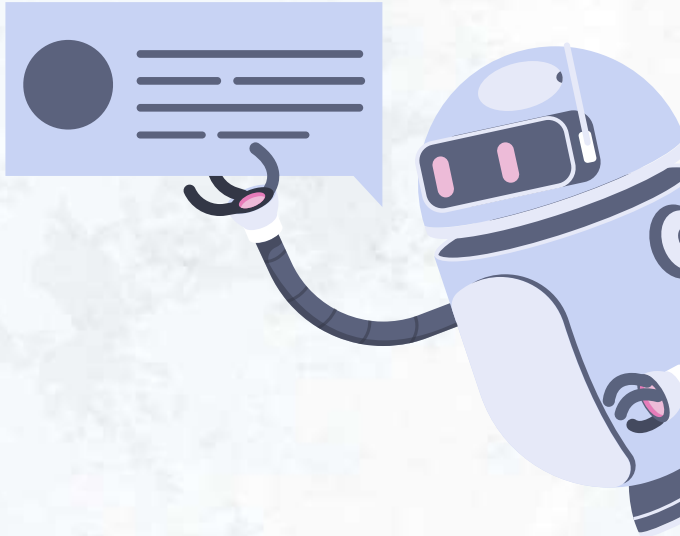
Section 1

01 →

Text preprocessing with Python.

What is NLTK?

NLTK stands for **Natural Language Toolkit**.



NLTK installation

- pip install nltk

NLTK data installation

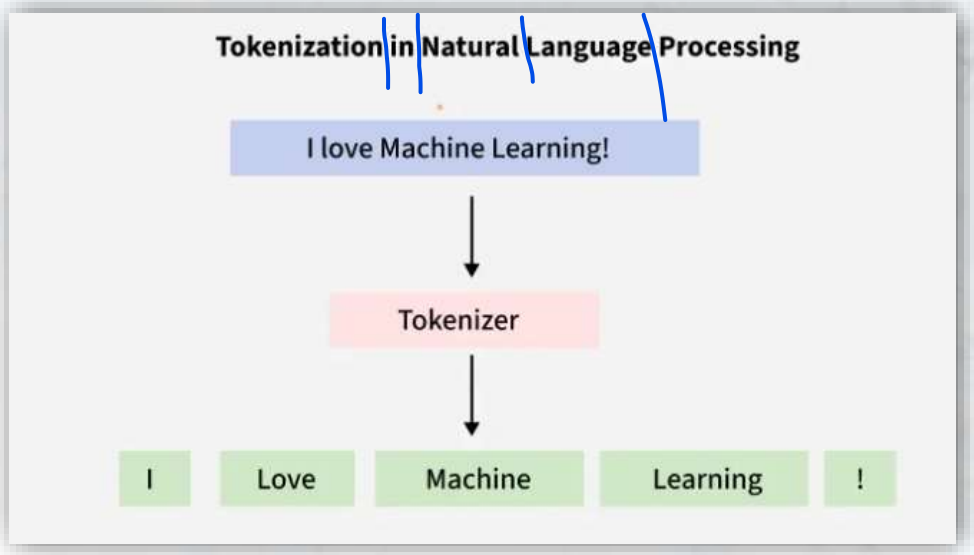
```
import nltk
nltk.download()
```

02 →

Tokeniztation

Tokenization:

Tokenization is the process of breaking down a text into smaller units called **tokens**.



Word Tokenization:

nltk.download('punkt')

```
import nltk
from nltk.tokenize import word_tokenize

text = "NLTK is a powerful NLP library."
tokens = word_tokenize(text)
print(tokens)
```

Output

['NLTK', 'is', 'a', 'powerful', 'NLP', 'library', '.']

Sentence Tokenization:

```
import nltk
from nltk.tokenize import sent_tokenize

text = "Hello there! I hope you're doing well. Have a great day!"
sentences = sent_tokenize(text)
print(sentences)
```

Output

['Hello there!', "I hope you're doing well.", 'Have a great day!']

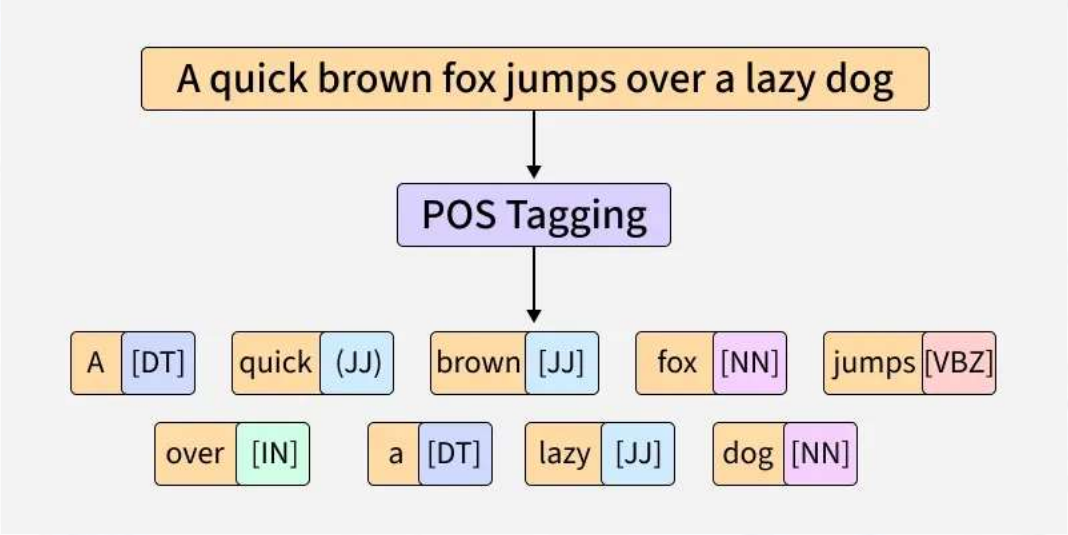
03 →

POS Tagging

POS Tagging

POS tagging stands for **Part-of-Speech tagging.**

It's the process of **assigning a part of speech (like noun, verb, adjective, etc.) to each word in a sentence** — based on both its definition and its context



Some POS Tags

POS Tag	Meaning	Example
DT	Determiner	The, a
JJ	Adjective	Tall , brown
NN	Noun	Cat , dog
VBZ	Verb (present tense)	jumps
IN	Preposition	over
PRP	Pronoun	She , he
RB	Adverb	Quickly

```python
import nltk
from nltk import word_tokenize, pos_tag

text = "The cat sat on the mat.
tokens = word_tokenize(text)
pos_tags = pos_tag(tokens)

print(pos_tags)
```

Output:

[('The', 'DT'), ('cat', 'NN'), ('sat', 'VBD'), ('on', 'IN'), ('the', 'DT'), ('mat', 'NN'), ('.', '.')]

04 →

NER
(Named Entity Recognition)

Named Entity Recognition

- **Named Entity Recognition (NER)** is the process of **identifying and classifying real-world entities** in text into predefined categories such as:

  PERSON → people's names
  ORG → organizations and companies
  GPE → countries, cities, or locations
  DATE → dates and time expressions

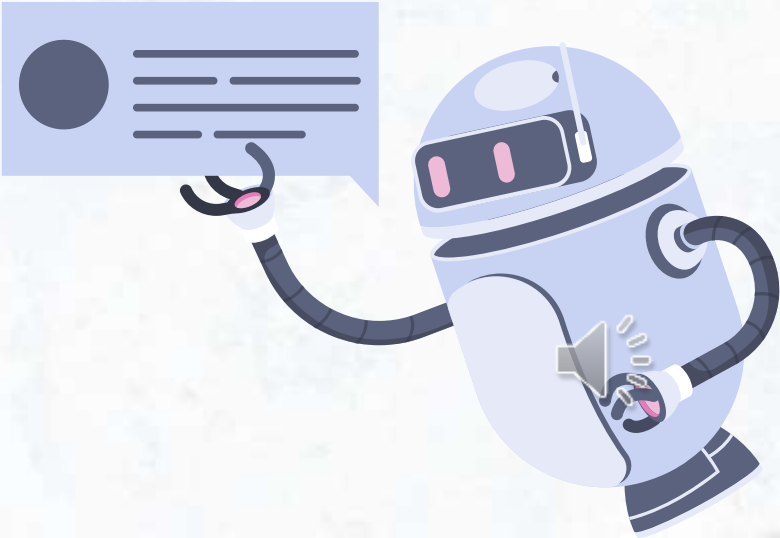
  MONEY → monetary values

  PRODUCT, EVENT, LANGUAGE, etc. (depending on the model)

John Doe **PERSON** has been working for Abcd Inc. **ORG** as a senior engineer since 2010 **DATE** in California **GPE** .

```python
import nltk
from nltk import word_tokenize, pos_tag, ne_chunk

# Sentence with a date
text = "John has been working for Abcd Inc. since 2010 in California."


# Step 1: Tokenize
tokens = word_tokenize(text)

# Step 2: POS tagging
pos_tags = pos_tag(tokens)

# Step 3: Named Entity Recognition
named_entities = ne_chunk(pos_tags)

print(named_entities)
```

Output

```
(S
  (PERSON John/NNP)
  has/VBZ
  been/VBN
  working/VBG
  for/IN
  (ORGANIZATION Abcd/NNP Inc./NNP)
  since/IN
  2010/CD
  in/IN
  (GPE California/NNP)
  ./.)
```

THANK YOU